

Pramo Jewels

Shipping Partner Integration Specification

Document ID: PJ-SHIP-001

Version: 1.0

Status: Approved for Development

Field	Value
Project	Pramo Jewels Premium E-Commerce Platform
Purpose	Define integration requirements for courier partners and end-to-end shipment m
Audience	Backend Developers, DevOps, Logistics Team, QA, Operations
Version	1.0

Core Shipping Modules

Integration Module	Specification
Courier Partners	Support Shippocket, Delhivery, Blue Dart, DTDC and future providers via adapter pattern.
Serviceability	PIN code validation before checkout.
Shipment Creation	Create shipment automatically after confirmed order.
AWB & Labels	Generate AWB numbers and printable shipping labels.
Pickup Scheduling	Book pickups through courier APIs.
Tracking	Real-time shipment status synchronization.
Returns (RTO)	Support return pickup and Return-to-Origin workflows.
Webhooks	Process courier events securely with signature verification.

Architecture

Use a provider abstraction layer so courier partners can be replaced or expanded without changing business logic.

Shipping Workflow

Order Confirmed → Serviceability Check → Courier Selection → Shipment Creation → AWB & Label Generation → Pickup → In Transit → Delivered → POD Recorded.

Courier Selection

Select the optimal courier using configurable rules such as delivery speed, shipping cost, destination, COD support and historical SLA performance.

Business Rules

Validate shipping addresses, support partial shipments, automatically update order status from courier events and maintain shipment history.

Security

Authenticate API requests, protect API keys in a secret manager, validate webhook signatures and encrypt sensitive shipment data in transit.

Monitoring

Track delivery success, SLA compliance, transit delays, RTO rate, pickup success, webhook failures and courier performance.

Integrations

Integrates with Orders, Inventory, Payments, Notifications, Customer Accounts, Analytics and Audit Logging.

Acceptance Criteria

Shipments are created reliably, tracking updates remain synchronized, courier failures are recoverable and all logistics events are audit logged.